

Parametric Inference

Two-Day Exam Revision Handout

Built from the 10 lecture notes (29 Jul – 28 Aug 2026) and Homeworks 1–3

0. The two-day plan

Your position: Lectures 1–5 (estimation) were studied but are stale; Lectures 6–10 (Bayes, minimax, testing, asymptotics) are the new material. The homeworks tell you the exam style: *short, self-contained derivations on named distributions*. Nothing is asked that needs more than one page of honest algebra. So the plan is built around **deriving, not reading**.

What the homeworks tell you about the exam

Fifteen problems across three sheets. Every single one has the form “here is a named family, derive the [UMVUE / Bayes estimate / minimax estimate / MP or UMP test].” There are **no** abstract measure-theoretic questions and **no** long asymptotic expansions. The families that recur: Binomial, Poisson, Exponential, Normal, Uniform, Cauchy, and power-series/negative-binomial. Master those seven and you have covered the observed question space.

Slot	Topic	What you actually do
Tonight (≈ 4 h) — rebuild the estimation core		
1.5 h	§1–§4: risk, unbiasedness, sufficiency, completeness, Rao–Blackwell, Lehmann–Scheffé	Read §1–§4 with a pen. Reproduce the factorisation theorem statement and the Lehmann–Scheffé proof from memory once.
1.0 h	§5–§7: exponential families, natural parameter space, ancillarity, Basu, Cramér–Rao	Write the exponential-family form for Bin, Poi, Exp, $N(\mu, \sigma^2)$, Gamma <i>from scratch</i> . Identify T , $c(\theta)$, natural space each time. Then learn the CRLB <i>attainment</i> condition in §7.
1.5 h	HW1 Q1–Q4, Q6	Attempt cold, 12 min each, then check the solutions PDF. These five are the highest-yield problems in the course.
Tomorrow morning (≈ 4 h) — Bayes and minimax (new)		
1.5 h	§8: prior/posterior, conjugacy, Bayes estimate = posterior mean, Bayes risk, improper priors	Derive all three conjugate pairs (Beta–Bin, Gamma–Poi, Normal–Normal) by hand. Do <i>not</i> memorise the posteriors; memorise the kernel-matching trick.
1.5 h	§9: minimax, constant-risk theorem, limits of Bayes rules	Reproduce the two-line proof of “Bayes + constant risk $\Rightarrow$ minimax” and the sequence version.
1.0 h	HW2 Q1–Q3	Cold attempt. Q1 and Q2 are the two canonical minimax arguments; expect one of them or a close cousin.
Tomorrow afternoon/evening (≈ 5 h) — testing (new, heaviest)		
2.0 h	§10: test functions, size/level/power, Neyman–Pearson with randomisation	Learn the NP statement exactly. Practise the loop: form $f_1/f_0 \rightarrow$ reduce to a statistic $\rightarrow$ invert to a region $\rightarrow$ set the size.
1.5 h	§11: MLR, Karlin–Rubin, UMP for one-sided hypotheses	Prove exponential family $\Rightarrow$ MLR. Know why two-sided UMP usually fails, and why $U[0, \theta]$ is the exception.
1.5 h	HW3 (all four) + HW2 Q5	These five are <i>exactly</i> the testing questions you will face. Do them under time.
Exam morning (≈ 2 h) — consolidate		
0.5 h	§12: consistency, b_n -consistency, MLE consistency	Light read; the cheapest section to score on.
1.0 h	§13–§14: the cheat sheets	Re-derive every row of the distribution table from the exponential-family form. If you can rebuild the table, you are ready.
0.5 h	Skim your own notes 1–10	Only to catch anything idiosyncratic the professor stressed.

Triage — if you run short of time

Drop, in this order: the Cauchy/ancillarity fine print (§6), the simple-linear-regression consistency (§12), and the multiparameter completeness detail (§5). **Never** drop: Lehmann–Scheffé (§4), exponential-family completeness (§5), Neyman–Pearson (§10), Karlin–Rubin (§11). Those four carry the paper.

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1 Decision-theoretic setup (Lec 1)

$X \sim P_\theta$, $\theta \in \Theta$. An *estimator* $T = T(X)$ of $\psi(\theta)$ is judged by its *risk* under squared error:

$$R(\theta, T) = \mathbb{E}_\theta[(T - \psi(\theta))^2] = \text{MSE}_\theta(T) = \underbrace{(\mathbb{E}_\theta T - \psi(\theta))^2}_{\text{bias}^2} + \text{Var}_\theta(T).$$

- T is **unbiased** for ψ if $\mathbb{E}_\theta T = \psi(\theta)$ for *all* θ .
- T^* is **UMVUE** if it is unbiased and $\text{Var}_\theta(T^*) \leq \text{Var}_\theta(T)$ for every unbiased T and every θ .
- T is **inadmissible** if some T' has $R(\theta, T') \leq R(\theta, T)$ for all θ , with strict inequality somewhere.

The pitfall Lecture 1 opens with

There is *no* uniformly best estimator without restricting the class. For any fixed θ_0 , the silly estimator $T \equiv \theta_0$ has $R(\theta_0, T) = 0$, so no T^* can beat everything everywhere. That is exactly *why* we impose unbiasedness (giving UMVUE), or average the risk (giving Bayes), or look at the worst case (giving minimax). Be able to say this in one sentence — it is a standard opening bookwork part.

Worked example (Lec 1, and a classic exam part). $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$. Among estimators $T_c = c \sum_i (X_i - \bar{X})^2$ of σ^2 , minimise the MSE. Write $W = \sum (X_i - \bar{X})^2$, so $W/\sigma^2 \sim \chi_{n-1}^2$, $\mathbb{E}W = (n-1)\sigma^2$, $\text{Var } W = 2(n-1)\sigma^4$, hence $\mathbb{E}W^2 = \text{Var } W + (\mathbb{E}W)^2 = \sigma^4(n^2 - 1)$. Then

$$R(\sigma^2, T_c) = \mathbb{E}(cW - \sigma^2)^2 = c^2 \mathbb{E}W^2 - 2c\sigma^2 \mathbb{E}W + \sigma^4 = \sigma^4 [c^2(n^2 - 1) - 2c(n - 1) + 1].$$

Minimising the quadratic in c gives $c^* = \frac{n-1}{n^2-1} = \frac{1}{n+1}$.

So $\frac{1}{n+1} \sum (X_i - \bar{X})^2$ beats the unbiased $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ in MSE at every σ^2 : **the UMVUE is inadmissible**. Memorise the constant $1/(n+1)$.

2 Does an unbiased estimator even exist? (Lec 2)

Before hunting for a UMVUE, ask whether ψ is *estimable*. The method is always the same: write $\mathbb{E}_\theta T = \psi(\theta)$ as an identity in θ , then use uniqueness of a series or transform expansion.

The three standard arguments

- Polynomial argument (Binomial).** $X \sim \text{Bin}(n, \theta)$, $\theta \in (0, 1)$. Then $\mathbb{E}_\theta T = \sum_{i=0}^n T(i) \binom{n}{i} \theta^i (1-\theta)^{n-i}$ is a *polynomial in θ of degree $\leq n$* . Hence ψ is unbiasedly estimable **iff** ψ is a polynomial of degree $\leq n$. In particular $\psi(\theta) = 1/\theta$ (or $(1-\theta)/\theta$) has *no* unbiased estimator: a degree- $\leq n$ polynomial cannot equal $1/\theta$ on an interval.
- Power-series argument (Poisson).** $X \sim \text{Poi}(\theta)$. Unbiasedness $\iff \sum_{i \geq 0} \frac{T(i)}{i!} \theta^i = \psi(\theta)e^\theta$ for all $\theta > 0$. So ψ is estimable **iff** $\psi(\theta)e^\theta$ is analytic on $(0, \infty)$, i.e. **iff** ψ is analytic; and then $T(i) = i! \times [i\text{-th Taylor coefficient of } \psi(\theta)e^\theta]$ is *unique*.
- Laplace-transform argument (Exponential).** $X \sim \frac{1}{\theta} e^{-x/\theta}$, $x > 0$. Then $\mathbb{E}_\theta T = \theta \iff \int_0^\infty T(x) e^{-x/\theta} dx = \theta^2$ for all $\theta > 0$. By uniqueness of the Laplace transform the solution T is unique (a.e.).

Remark. Whenever the unbiased estimator is *unique*, it is automatically the UMVUE — there is nothing to compare it with. This “uniqueness $\Rightarrow$ UMVUE” shortcut appears repeatedly and is worth one sentence in any answer.

Negative binomial (Lec 7). “Toss until 2 tails”: $f(x | \theta) = \binom{x-1}{1} \theta^{x-2} (1-\theta)^2$, $x \geq 2$. No unbiased estimator of θ exists (same polynomial-type obstruction). Contrast HW1 Q1, where θ is estimable for a differently parametrised negative binomial — estimability depends on the parametrisation, not on the distribution.

Cauchy (Lec 7). For $X \sim \frac{1}{\pi\{1+(x-\theta)^2\}}$ with $n = 1$ there is no unbiased estimator of θ ; X itself has no mean.

3 Sufficiency (Lec 2–3)

Definition 3.1. $T(X)$ is **sufficient** for θ if the conditional distribution of X given $T = t$ does not depend on θ .

Theorem 3.2 (Neyman–Fisher factorisation). T is sufficient $\iff$ the joint pdf/pmf factorises as

$$f(\mathbf{x} | \theta) = g(\theta, T(\mathbf{x})) \cdot h(\mathbf{x}),$$

with h free of θ and the θ -dependence entering only through T .

Model (n iid)	Sufficient statistic	Note
$\text{Bin}(m, \theta)$	$\sum X_i$	complete
$\text{Poi}(\theta)$	$\sum X_i$	complete; $\sum X_i \sim \text{Poi}(n\theta)$
$\text{Exp}(\text{mean } \theta)$	$\sum X_i$	complete; $\sum X_i \sim \text{Gamma}(n, \theta)$
$N(\theta, 1)$	$\bar{X}$	complete
$N(\mu, \sigma^2)$	$(\sum X_i, \sum X_i^2)$	complete (2-param. exp. family)
$U[0, \theta]$	$X_{(n)}$	complete
$U[\phi, \psi]$	$(X_{(1)}, X_{(n)})$	complete
$U[\theta, \theta + 1]$	$(X_{(1)}, X_{(n)})$	sufficient, not complete
$\text{Cauchy}(\theta)$	full order statistic	sufficient, not complete

Indicator bookkeeping for the uniform families. This is where marks are lost. For $U[\phi, \psi]$,

$$f(\mathbf{x} | \phi, \psi) = (\psi - \phi)^{-n} \prod_i \mathbb{I}[\phi < x_i < \psi] = (\psi - \phi)^{-n} \mathbb{I}[\phi < x_{(1)}] \cdot \mathbb{I}[x_{(n)} < \psi],$$

because “all $x_i > \phi$ ” $\iff$ “ $x_{(1)} > \phi$ ”. Always collapse the product of indicators into the two extreme order statistics before quoting the factorisation theorem.

Two-point parameter space (Lec 5). If $\Theta = \{\theta_0, \theta_1\}$, the likelihood ratio $T(x) = f_{\theta_1}(x)/f_{\theta_0}(x)$ is **sufficient**: take $h(x) = f_{\theta_0}(x)$ and $g(\theta, T) = T$ if $\theta = \theta_1$, $= 1$ if $\theta = \theta_0$. This is the bridge from estimation to testing — it is exactly why the Neyman–Pearson test depends on the data only through the likelihood ratio.

4 Completeness, Rao–Blackwell, Lehmann–Scheffé (Lec 2–3)

Definition 4.1. A sufficient statistic T is **complete** if for every function g ,

$$\mathbb{E}_\theta g(T) = 0 \quad \forall \theta \in \Theta \implies g(T) = 0 \text{ a.s. for all } \theta.$$

Why completeness is the right notion

If T is complete sufficient and $h_1(T), h_2(T)$ are both unbiased for ψ , then $\mathbb{E}_\theta[h_2(T) - h_1(T)] = 0$ for all θ , so $h_1 \equiv h_2$. There is at most one unbiased estimator that is a function of a complete sufficient statistic. Uniqueness is the whole point.

Theorem 4.2 (Rao–Blackwell). Let T be any unbiased estimator of $\psi(\theta)$ and S sufficient. Put $T^* = \mathbb{E}[T | S]$. Then T^* is a statistic (sufficiency makes it free of θ), $\mathbb{E}_\theta T^* = \psi(\theta)$, and $R(\theta, T^*) \leq R(\theta, T)$ for all θ .

Proof (memorise — it is three lines). $\mathbb{E} T^* = \mathbb{E} \mathbb{E}[T | S] = \mathbb{E} T = \psi(\theta)$, so the bias is unchanged. By the variance decomposition,

$$\text{Var}(T) = \mathbb{E}[\text{Var}(T | S)] + \text{Var}(\mathbb{E}[T | S]) \geq \text{Var}(T^*),$$

since the first term is ≥ 0 . Equal bias plus smaller variance gives smaller MSE. Equality holds iff $\text{Var}(T | S) = 0$ a.s., i.e. T is already a function of S . $\square$

Theorem 4.3 (Lehmann–Scheffé). If S is **complete** and sufficient and $h(S)$ is unbiased for $\psi(\theta)$, then $h(S)$ is the **UMVUE** of $\psi(\theta)$, and it is a.s. unique.

Proof. Let T be any unbiased estimator. Rao–Blackwellise: $T^* = \mathbb{E}[T | S]$ is unbiased, a function of S , and $\text{Var}(T^*) \leq \text{Var}(T)$. By completeness $T^* = h(S)$ a.s. Hence $\text{Var}(h(S)) \leq \text{Var}(T)$ for every unbiased T . $\square$

The UMVUE recipe — two routes, use whichever is shorter

Route A (guess-and-check). Find a complete sufficient S ; then find *any* function h with $\mathbb{E}_\theta h(S) = \psi(\theta)$, typically by expanding both sides in a series and matching coefficients. Done. (Used in HW1 Q1, Q2.)

Route B (Rao–Blackwellise). Find a crude unbiased estimator T_0 — often an indicator like $\mathbb{I}[X_1 > 0]$ or $\mathbb{I}[X_1 = 0]$ — and compute $\mathbb{E}[T_0 | S]$. The conditional distribution of X_1 given S is the only real work. (Used in HW1 Q3.)

Standard UMVUEs to know cold.

- $U[0, \theta]$: $X_{(n)}$ is complete sufficient with $\mathbb{E} X_{(n)} = \frac{n}{n+1} \theta$, so the UMVUE of θ is $\frac{n+1}{n} X_{(n)}$.
- $\text{Poi}(\theta)$, one observation: UMVUE of $e^{c\theta}$ is $(1+c)^X$; with n observations, $(1+c/n)^S$ where $S = \sum X_i$.
- $N(\theta, 1)$: UMVUE of $\Phi(\theta) = \mathbb{P}(X_1 > 0)$ is $\Phi(\bar{X} \sqrt{n/(n-1)})$.
- $\text{Bin}(n, \theta)$: UMVUE of θ is X/n ; UMVUE of $\theta(1-\theta)$ is $\frac{X(n-X)}{n(n-1)}$.

5 Exponential families (Lec 4–5)

Definition 5.1 (One-parameter). $f(x | \theta) = p(\theta) h(x) \exp\{c(\theta) T(x)\}$. For $X_1, \dots, X_n$ iid,

$$f(\mathbf{x} | \theta) = p(\theta)^n \left(\prod_i h(x_i) \right) \exp \left\{ c(\theta) \sum_i T(x_i) \right\},$$

so by the factorisation theorem $S = \sum_i T(X_i)$ is sufficient. Call $\eta = c(\theta)$ the **natural parameter** and $\Gamma = \{c(\theta) : \theta \in \Theta\}$ the **natural parameter space**.

Theorem 5.2 (Completeness criterion — the most used result in the course). If the natural parameter space Γ contains an open interval (an interval with non-empty interior), then $S = \sum_i T(X_i)$ is **complete sufficient**.

Multiparameter version (Lec 5): if $f(x | \theta) = p(\theta)h(x) \exp\{\sum_{j=1}^k c_j(\theta)T_j(x)\}$ with $\theta = (\theta_1, \dots, \theta_k)$, then $S = (\sum_i T_1(X_i), \dots, \sum_i T_k(X_i))$ is sufficient, and complete provided the natural parameter space $\{(c_1(\theta), \dots, c_k(\theta)) : \theta \in \Theta\}$ contains a k -dimensional rectangle with non-empty interior.

Family	$T(x)$	$c(\theta)$	natural space Γ	complete?
$\text{Bin}(m, \theta), \theta \in (0, 1)$	x	$\log \frac{\theta}{1-\theta}$	$\mathbb{R}$	yes
$\text{Poi}(\theta), \theta > 0$	x	$\log \theta$	$\mathbb{R}$	yes
$\text{Exp}(\text{mean } \theta)$	x	$-1/\theta$	$(-\infty, 0)$	yes
$N(\theta, 1)$	x	θ	$\mathbb{R}$	yes
$N(\mu, \sigma^2)$	(x, x^2)	$(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2})$	$\mathbb{R} \times (-\infty, 0)$	yes
$\text{Gamma}(\alpha, \beta)$	$(\log x, x)$	$(\alpha - 1, -1/\beta)$	open rectangle	yes
$U[0, \theta]$	<i>not</i> an exponential family (support depends on θ)			yes, shown directly
$\text{Cauchy}(\theta)$	<i>not</i> an exponential family			no

Where the criterion bites

$N(\theta, \theta^2)$ has natural parameters $(\frac{1}{\theta}, -\frac{1}{2\theta^2})$, which trace a *curve* in $\mathbb{R}^2$, not a rectangle. The criterion does not apply and completeness fails. Likewise $U[\theta, \theta + 1]$ is not an exponential family at all, and its sufficient statistic $(X_{(1)}, X_{(n)})$ is not complete because $X_{(n)} - X_{(1)}$ is ancillary. The phrase to write is: “the natural parameter space does not contain a rectangle with non-empty interior, so the criterion is unavailable.”

Power-series family (Lec 4).

$$f(x | \theta) = \frac{a_x \theta^x}{A(\theta)}, \quad x = 0, 1, 2, \dots, \quad A(\theta) = \sum_{x \geq 0} a_x \theta^x, \quad \theta \in (0, b), \quad a_x \geq 0.$$

This is an exponential family with $T(x) = x$, $c(\theta) = \log \theta$, natural space $(-\infty, \log b)$ — an open interval, so X (and $\sum X_i$) is complete sufficient. Two identities worth memorising, with $\psi(\theta) := \mathbb{E}_\theta X$:

$$\mathbb{E}_\theta X = \frac{\theta A'(\theta)}{A(\theta)} = \psi(\theta), \quad \text{Var}_\theta X = \theta \psi'(\theta).$$

Bernoulli, Binomial, Poisson, Geometric and Negative Binomial are all power-series families. (HW1 Q4 is exactly this; see the solutions PDF.)

6 Ancillarity and Basu’s theorem (Lec 4)

Definition 6.1. $V(X)$ is **ancillary** if its distribution does not depend on θ .

Theorem 6.2 (Basu). If S is complete sufficient for θ and V is ancillary, then $S \perp\!\!\!\perp V$ under every P_θ .

How to spot an ancillary statistic in one line

Location family $f(x - \theta)$: any *difference* is ancillary — $X_i - X_j$, $X_{(n)} - X_{(1)}$, S^2 , the whole vector $(X_1 - \bar{X}, \dots, X_n - \bar{X})$.

Scale family $\frac{1}{\theta} f(x/\theta)$: any *ratio* is ancillary — X_i/X_j , $X_{(n)}/X_{(1)}$, $\sum_{j \leq p} X_j / \sum_{k \leq q} X_k$, $\bar{X}/S$.

Proof template: write $X_i = \theta Z_i$ (scale) or $X_i = \theta + Z_i$ (location) with Z_i free of θ ; the statistic is unchanged, hence its law is free of θ .

Cauchy: sufficiency without completeness. For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Cauchy}(\theta)$ the full order statistic $S = (X_{(1)}, \dots, X_{(n)})$ is sufficient (and minimal), but *not* complete: for bounded g , the statistic $g(X_{(i)} - X_{(j)})$ has a θ -free distribution, so with $c = \mathbb{E}g(X_{(i)} - X_{(j)})$ we get $\mathbb{E}_\theta[g(X_{(i)} - X_{(j)}) - c] = 0$ for all θ while $g - c \not\equiv 0$. This is the standard counterexample; know it.

The two textbook Basu applications.

- (i) $N(\mu, \sigma^2)$ with σ^2 known: $\bar{X}$ is complete sufficient for μ , and S^2 is ancillary, hence $\bar{X} \perp\!\!\!\perp S^2$. (The slick proof of the classical independence.)
- (ii) $\text{Exp}(\theta)$: $\sum X_i$ is complete sufficient, $\sum_{j \leq p} X_j / \sum_{k \leq q} X_k$ is scale-invariant hence ancillary, so the two are independent. (HW1 Q6.)

7 Information inequality / Cramér–Rao (Lec 2, HW1 Q4)

Under regularity (support free of θ , differentiation under the integral sign legal), the **Fisher information** in one observation is

$$I(\theta) = \text{Var}_\theta \left(\frac{\partial}{\partial \theta} \log f(X | \theta) \right) = \mathbb{E}_\theta \left[\left(\frac{\partial}{\partial \theta} \log f \right)^2 \right] = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f \right],$$

and for n iid observations $I_n(\theta) = nI(\theta)$.

Theorem 7.1 (CRLB). For any T with $\mathbb{E}_\theta T = \psi(\theta)$, $\text{Var}_\theta(T) \geq \frac{[\psi'(\theta)]^2}{n I(\theta)}$.

Attainment — the exam-relevant half

Equality holds at every θ **iff** the score factorises as

$$\frac{\partial}{\partial \theta} \log f(\mathbf{x} | \theta) = k(\theta) [T(\mathbf{x}) - \psi(\theta)]$$

for some $k(\theta)$, i.e. **iff the model is a one-parameter exponential family and ψ is the mean of its natural sufficient statistic**. Consequences:

- $\bar{X}$ attains the CRLB for μ in $N(\mu, \sigma_0^2)$, for θ in $\text{Poi}(\theta)$, for θ in $\text{Exp}(\theta)$, and for $\mathbb{E}X$ in any power-series family. (The equality condition in Cauchy–Schwarz is the source.)
- The CRLB is *not* attained for e.g. σ^2 in $N(\mu, \sigma^2)$, or for non-linear ψ . A UMVUE can therefore exist with variance strictly above the CRLB — so “does not attain CRLB” never disproves UMVUE-hood.
- The CRLB is **meaningless** for $U[0, \theta]$: the support depends on θ , regularity fails, and indeed $\frac{n+1}{n} X_{(n)}$ has variance of order θ^2/n^2 , far below any bound you would naively compute.

Consistency of UMVUEs (Lec 4). If T_n is the UMVUE of $\psi(\theta)$ based on $X_1, \dots, X_n$ and there exists *some* unbiased sequence whose variance tends to 0, then $\text{Var}(T_n) \leq \text{Var}(\tilde{T}_n) \rightarrow 0$, and with zero bias Chebyshev gives $T_n \xrightarrow{p} \psi(\theta)$.

8 Bayes estimation (Lec 6–7)

Prior $\pi(\theta)$ on Θ ; data $X \sim f(x | \theta)$. The **posterior** is

$$\pi(\theta | x) = \frac{f(x | \theta)\pi(\theta)}{\int_{\Theta} f(x | \theta')\pi(\theta') d\theta'} \propto f(x | \theta) \pi(\theta).$$

Theorem 8.1 (Bayes estimate under squared error). *The estimator minimising the **Bayes risk** $r(\pi, T) = \int_{\Theta} R(\theta, T)\pi(\theta) d\theta = \mathbb{E}_{\pi}[R(\theta, T)]$ is the **posterior mean** $T_{\pi}(x) = \mathbb{E}[\theta | X = x]$.*

Proof (know this). Write $r(\pi, T) = \iint (T(x) - \theta)^2 f(x | \theta)\pi(\theta) dx d\theta = \int m(x) \left[\int (T(x) - \theta)^2 \pi(\theta | x) d\theta \right] dx$, where m is the marginal density of X . For each fixed x the inner bracket is minimised over the number $T(x)$ by the mean of $\pi(\cdot | x)$, since $a \mapsto \mathbb{E}[(a - \theta)^2 | x]$ is minimised at $a = \mathbb{E}[\theta | x]$. The minimum value is the posterior variance, so $r(\pi, T_{\pi}) = \mathbb{E}_m[\text{Var}(\theta | X)]$ — a formula you will use repeatedly in the minimax problems. $\square$

Conjugate families — derive, do not memorise

Drop all factors free of θ and match the kernel.

Model	Prior	Posterior	Bayes estimate
$X \sim \text{Bin}(n, \theta)$	$\text{Beta}(p, q)$	$\text{Beta}(p + x, q + n - x)$	$\frac{p + X}{p + q + n}$
$X \sim \text{Poi}(\theta)$	$\text{Gamma}(\alpha, \lambda)$	$\text{Gamma}(\alpha + x, \frac{\lambda}{1 + \lambda})$	$\frac{\lambda(\alpha + X)}{1 + \lambda}$
$X \sim N(\theta, \sigma^2), \sigma^2 \text{ known}$	$N(\mu, \tau^2)$	$N\left(\frac{\tau^2 x + \sigma^2 \mu}{\sigma^2 + \tau^2}, \frac{\sigma^2 \tau^2}{\sigma^2 + \tau^2}\right)$	$\frac{\tau^2 X + \sigma^2 \mu}{\sigma^2 + \tau^2}$
$X_{(n)} \text{ from } U[a, \theta]$	$\text{Pareto}(\alpha, a)$	$\text{Pareto}(\alpha + n, \max(a, x_{(n)}))$	$\frac{(\alpha + n) \max(a, X_{(n)})}{\alpha + n - 1}$

(Gamma parametrised by shape and *scale*.) For n iid observations replace x by the sufficient statistic: $\sum x_i$ for Bin/Poi, and $\bar{X}$ with $\sigma^2 \rightarrow \sigma^2/n$ for the Normal.

Three facts the professor emphasised.

1. **The posterior depends on the data only through a sufficient statistic.** If $f(x | \theta) = g(\theta, T(x))h(x)$ then $h(x)$ cancels between numerator and denominator: $\pi(\theta | x) = g(\theta, T(x))\pi(\theta) / \int g(\theta, T(x))\pi(\theta) d\theta$. Consequently *every Bayes and generalized Bayes estimate is a function of the sufficient statistic*.
2. **A Bayes estimate is essentially never unbiased.** Suppose T is both Bayes (so $T = \mathbb{E}[\theta | X]$) and unbiased (so $\mathbb{E}[T | \theta] = \theta$). Writing $\mathbb{E}_m$ for expectation under the joint law and conditioning two different ways,

$$\mathbb{E}_m[T\theta] = \mathbb{E}[\theta \mathbb{E}(T | \theta)] = \mathbb{E}[\theta^2], \quad \mathbb{E}_m[T\theta] = \mathbb{E}[T \mathbb{E}(\theta | X)] = \mathbb{E}[T^2].$$

Hence $\mathbb{E}_m[(T - \theta)^2] = \mathbb{E}[T^2] + \mathbb{E}[\theta^2] - 2\mathbb{E}_m[T\theta] = 0$, forcing $T = \theta$ a.s. — impossible except in degenerate cases. *Memorise this four-line argument.*

3. **Improper priors and generalized Bayes.** π need not integrate to 1. If the posterior is still a proper density, its mean is the *generalized Bayes estimate*. Examples: $X \sim N(\theta, 1)$ with $\pi \equiv 1$ gives $\pi(\theta | x) = N(x, 1)$ and estimate X ; with n iid observations, $\bar{X}$. For $U[\theta - \frac{1}{2}, \theta + \frac{1}{2}]$ with $\pi \equiv 1$ the posterior is uniform on $[x_{(n)} - \frac{1}{2}, x_{(1)} + \frac{1}{2}]$ and the estimate is the midrange $\frac{1}{2}(X_{(1)} + X_{(n)})$ (HW2 Q3).

9 Minimax estimation (Lec 8–9)

Definition 9.1. T^* is **minimax** if $\sup_{\theta \in \Theta} R(\theta, T^*) = \inf_T \sup_{\theta \in \Theta} R(\theta, T)$. An estimator whose risk is constant in θ is called an **equalizer**.

Theorem 9.2 (Constant-risk criterion). If T_π is Bayes w.r.t. π and $R(\theta, T_\pi) = c$ is constant in θ , then T_π is minimax (and π is a least favourable prior).

Proof. For any T : $\sup_{\theta} R(\theta, T) \geq r(\pi, T) \geq r(\pi, T_\pi) = c = \sup_{\theta} R(\theta, T_\pi)$. The first step is “ $\sup \geq$ average”, the second is Bayes optimality. $\square$

Theorem 9.3 (Limit of Bayes rules — the workhorse). Let $\{\pi_k\}$ be priors with Bayes risks $r(\pi_k, T_{\pi_k}) \rightarrow c$. If T^* satisfies $\sup_{\theta} R(\theta, T^*) = c$, then T^* is minimax.

Proof. For any T and every k : $\sup_{\theta} R(\theta, T) \geq r(\pi_k, T) \geq r(\pi_k, T_{\pi_k})$. Let $k \rightarrow \infty$ to get $\sup_{\theta} R(\theta, T) \geq c = \sup_{\theta} R(\theta, T^*)$. $\square$

The three canonical minimax results

1. $X \sim \text{Bin}(n, \theta)$. With the prior $\text{Beta}(\frac{\sqrt{n}}{2}, \frac{\sqrt{n}}{2})$ the Bayes estimate is

$$T^* = \frac{X + \sqrt{n}/2}{n + \sqrt{n}},$$

whose risk is the *constant* $\frac{1}{4(1 + \sqrt{n})^2}$; hence T^* is minimax. By contrast the UMVUE

X/n has $\sup_{\theta} R = \frac{1}{4n}$, which is larger — **the UMVUE is not minimax**.

2. $X \sim N(\theta, 1)$, $\theta \in \mathbb{R}$. X is minimax with value 1. Use $\pi_k = N(0, k)$: the Bayes estimate is $kX/(k+1)$ with Bayes risk $\frac{k}{k+1} \rightarrow 1 = \sup_{\theta} R(\theta, X)$; apply the limit theorem. (HW2 Q1.)
3. $X \sim \text{Poi}(\lambda)$, $\lambda \in (0, \infty)$. Every estimator has infinite maximum risk. Use $\pi_k = \text{Gamma}(1, k)$, whose Bayes risk is $k^2/(1+k) \rightarrow \infty$. An unbounded parameter space with unbounded variance kills the problem. (HW2 Q2.) The same happens for $U[a, \theta]$ with θ unbounded.

10 Testing: setup and Neyman–Pearson (Lec 6, 9)

$H_0 : \theta \in \Theta_0$ vs $H_A : \theta \in \Theta_A$, $\Theta = \Theta_0 \cup \Theta_A$ disjointly.

Definition 10.1. A **test function** is $\phi : \mathcal{X} \rightarrow [0, 1]$; $\phi(x)$ is the probability of rejecting H_0 when $X = x$. Its **power function** is $\beta_{\phi}(\theta) = \mathbb{E}_{\theta}\phi(X)$. The **size** is $\sup_{\theta \in \Theta_0} \beta_{\phi}(\theta)$; ϕ has **level** α if its size is $\leq \alpha$. Type I error = $\beta_{\phi}(\theta)$ for $\theta \in \Theta_0$; Type II error = $1 - \beta_{\phi}(\theta)$ for $\theta \in \Theta_A$.

Sufficiency reduces the problem (Lec 6). If T is sufficient and ϕ is any test, then $\psi(T) := \mathbb{E}[\phi(X) | T]$ is a legitimate test function (free of θ by sufficiency) with $\mathbb{E}_{\theta}\psi = \mathbb{E}_{\theta}\phi$ for all θ — identical size and power. So it is enough to search among tests depending only on a sufficient statistic.

Theorem 10.2 (Neyman–Pearson lemma). Test $H_0 : X \sim f_0$ against $H_A : X \sim f_1$ (both simple). Fix $\alpha \in (0, 1)$.

(i) (**Existence**) There exist $k \geq 0$ and $\gamma \in [0, 1]$ such that

$$\phi^*(x) = \begin{cases} 1, & f_1(x) > k f_0(x), \\ \gamma, & f_1(x) = k f_0(x), \\ 0, & f_1(x) < k f_0(x), \end{cases} \quad \mathbb{E}_0 \phi^* = \alpha.$$

- (ii) (**Sufficiency**) Any such ϕ^* is most powerful (MP) of its size.
 (iii) (**Necessity**) Any MP level- α test has this form a.e., except possibly on $\{f_1 = kf_0\}$.

Proof of (ii) — short and often asked. Let ϕ be any test with $\mathbb{E}_0\phi \leq \alpha = \mathbb{E}_0\phi^*$. On $\{f_1 > kf_0\}$ we have $\phi^* = 1 \geq \phi$; on $\{f_1 < kf_0\}$ we have $\phi^* = 0 \leq \phi$. In both cases $(\phi^* - \phi)(f_1 - kf_0) \geq 0$, and this holds trivially on $\{f_1 = kf_0\}$ too. Integrating,

$$0 \leq \int (\phi^* - \phi)(f_1 - kf_0) = (\mathbb{E}_1\phi^* - \mathbb{E}_1\phi) - k(\mathbb{E}_0\phi^* - \mathbb{E}_0\phi) \leq \mathbb{E}_1\phi^* - \mathbb{E}_1\phi,$$

since $\mathbb{E}_0\phi^* - \mathbb{E}_0\phi \geq 0$ and $k \geq 0$. Hence $\mathbb{E}_1\phi^* \geq \mathbb{E}_1\phi$: ϕ^* is MP. $\square$

The NP algorithm — follow it mechanically

1. Write $\Lambda(x) = f_1(x)/f_0(x)$ and **simplify until only a statistic remains**.
2. Determine the shape of $\{\Lambda > k\}$: is it a one-sided region in that statistic (usual), or an interval, or a union of two tails? (Λ need not be monotone — see HW2 Q5.)
3. Re-express the region in terms of a statistic with a *known null distribution*.
4. Solve $\mathbb{P}_0(\text{region}) = \alpha$ for the cutoff. If the null distribution is **discrete**, the exact size is generally unattainable, so **randomise**: take the smallest c with $\mathbb{P}_0(T > c) \leq \alpha$ and set $\gamma = \frac{\alpha - \mathbb{P}_0(T > c)}{\mathbb{P}_0(T = c)}$.
5. State the power if asked.

Support matters. If $f_1 > 0$ where $f_0 = 0$, the ratio is $+\infty$ there and the MP test *must* reject on that set — this costs nothing in size and gains power. That is the entire mechanism behind the $U[0, \theta]$ problem (HW3 Q4).

11 MLR and UMP tests (Lec 9)

Definition 11.1. $\{f_\theta : \theta \in (a, b)\}$ has **monotone likelihood ratio (MLR)** in $T(x)$ if for every $\theta_0 < \theta_1$ the ratio $f_{\theta_1}(x)/f_{\theta_0}(x)$ is a non-decreasing function of $T(x)$.

Theorem 11.2. The one-parameter exponential family $f(x | \theta) = p(\theta)h(x)e^{c(\theta)T(x)}$ has MLR in $T(x)$ provided $c(\theta)$ is non-decreasing in θ .

Proof. $\frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} = \frac{p(\theta_1)}{p(\theta_0)} \exp\{[c(\theta_1) - c(\theta_0)]T(x)\}$, and $c(\theta_1) - c(\theta_0) \geq 0$, so the ratio is a non-decreasing function of $T(x)$. $\square$

Theorem 11.3 (Karlin–Rubin). If the family has MLR in T , then for $H_0 : \theta \leq \theta_0$ vs $H_A : \theta > \theta_0$ the test

$$\phi^*(x) = \begin{cases} 1, & T(x) > c \\ \gamma, & T(x) = c \\ 0, & T(x) < c \end{cases} \quad \text{with } \mathbb{E}_{\theta_0}\phi^* = \alpha$$

is **UMP** of level α . Moreover $\beta_{\phi^*}(\theta)$ is non-decreasing in θ , so the size is attained at $\theta = \theta_0$.

Why this works — the sentence to write in the exam

Fix any $\theta_1 > \theta_0$. By Neyman–Pearson, the MP test of the *simple* $H_0 : \theta = \theta_0$ against $H_A : \theta = \theta_1$ rejects for large $f_{\theta_1}/f_{\theta_0}$, i.e. for large T by MLR. **The cutoff c depends only on α and the null distribution, not on θ_1** — so the same test is MP against every $\theta_1 > \theta_0$ simultaneously, hence UMP. The monotone power function then upgrades $H_0 : \theta = \theta_0$ to

$H_0 : \theta \leq \theta_0$ without changing the size.

MLR status of the standard families. Binomial (in x), Poisson (in $\sum x_i$), Exponential (in $\sum x_i$), $N(\theta, 1)$ (in $\bar{X}$) and $U[0, \theta]$ (in $x_{(n)}$, in the generalised sense) all have MLR. **Cauchy does not:** $f_{\theta_1}/f_{\theta_0}$ is a ratio of two quadratics, which tends to 1 as $x \rightarrow \pm\infty$ in both directions, so it cannot be monotone.

Two-sided alternatives

For $H_0 : \theta = \theta_0$ vs $H_A : \theta \neq \theta_0$ in an MLR family, no UMP test exists: the best test against $\theta_1 > \theta_0$ rejects for large T , the best against $\theta_1 < \theta_0$ rejects for small T , and no single test does both. One then restricts to unbiased tests (UMPU). **Exception:** in non-regular families where the support moves with θ , a two-sided UMP test can exist — $U[0, \theta]$ with $H_0 : \theta = 1$ is the standard example (HW3 Q4), because rejecting on $\{X > 1\}$ is free under H_0 .

12 Consistency and asymptotics (Lec 4, 10)

Definition 12.1. T_n is (weakly) **consistent** for $\psi(\theta)$ if $T_n \xrightarrow{p} \psi(\theta)$ under P_θ for every θ . For $b_n \uparrow \infty$, T_n is **b_n -consistent** if $b_n(T_n - \theta) = O_p(1)$, i.e. $T_n - \theta = O_p(1/b_n)$.

Sufficient condition: if $\mathbb{E}_\theta T_n \rightarrow \psi(\theta)$ and $\text{Var}_\theta(T_n) \rightarrow 0$ then T_n is consistent (Chebyshev). This handles almost every exam instance.

Sample median (Lec 10). Let $F(\theta) = \frac{1}{2}$ with F strictly increasing and continuous near θ , $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$, and $\tilde{X}_n$ the sample median. Then

$$\mathbb{P}(\tilde{X}_n - \theta > \varepsilon) = \mathbb{P}(\text{at least half the } X_i > \theta + \varepsilon) = \mathbb{P}(\text{Bin}(n, 1 - F(\theta + \varepsilon)) \geq n/2) \rightarrow 0,$$

since $1 - F(\theta + \varepsilon) < \frac{1}{2}$; symmetrically for the other tail. Hence $\tilde{X}_n \xrightarrow{p} \theta$. This is the right starting estimator for the Cauchy location problem, because $\bar{X}$ is not even consistent there — $\bar{X}$ is itself Cauchy(θ) for every n .

MLE. $\hat{\theta}_n = \arg \max_\theta \prod_i f(X_i | \theta)$.

- Consistent for Binomial, Poisson, Exponential and Normal, and generally whenever the usual regularity conditions hold and the likelihood equation has a unique root for each n .
- $U[0, \theta]$: $\hat{\theta}_n = X_{(n)}$, with $\mathbb{E}X_{(n)} = \frac{n}{n+1}\theta$ (biased) but $\mathbb{P}(\hat{\theta}_n > \theta + \varepsilon) = 0$ and $\mathbb{P}(\hat{\theta}_n < \theta - \varepsilon) = \left(\frac{\theta - \varepsilon}{\theta}\right)^n \rightarrow 0$, so consistent — and in fact n -consistent, not $\sqrt{n}$ -consistent.
- Cauchy: the likelihood equation $\sum_i \frac{2(x_i - \theta)}{1 + (x_i - \theta)^2} = 0$ is a polynomial equation with no closed form and possibly several roots; solve numerically starting from the sample median (HW2 Q4).

Simple linear regression (Lec 10). $Y_i = \alpha + \beta x_i + \varepsilon_i$, $\varepsilon_i \stackrel{\text{iid}}{\sim}$ with mean 0, variance σ^2 .

$$\hat{\beta}_{LS} = \frac{\sum_i (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_i (x_i - \bar{x})^2}, \quad \mathbb{E}\hat{\beta}_{LS} = \beta, \quad \text{Var}(\hat{\beta}_{LS}) = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}.$$

So $\hat{\beta}_{LS}$ is consistent **iff** $\sum_i (x_i - \bar{x})^2 \rightarrow \infty$: the design points must be sufficiently spread out. And $\hat{\alpha}_{LS} = \bar{Y} - \hat{\beta}_{LS}\bar{x}$ is consistent provided additionally $\bar{x}$ converges to a constant.

13 Distribution cheat sheet

Model (n iid)	Complete S	suff.	UMVUE of θ	Bayes (conj. prior)	UMP for $\theta \leq \theta_0$ vs $> \theta_0$
Bern(θ)	$\sum X_i$		$\bar{X}$	$\frac{p+S}{p+q+n}$ (Beta)	reject $S > c$, randomise
Poi(θ)	$\sum X_i$		$\bar{X}$	$\frac{\lambda(\alpha+S)}{1+n\lambda}$ (Gamma)	reject $S > c$, randomise
Exp(θ)	$\sum X_i$		$\bar{X}$	inverse-Gamma	reject $2S/\theta_0 > \chi_{2n,\alpha}^2$
$N(\theta, 1)$	$\bar{X}$		$\bar{X}$	$\frac{\tau^2 \bar{X} + (\sigma^2/n)\mu}{\sigma^2/n + \tau^2}$	reject $\bar{X} > \theta_0 + z_\alpha/\sqrt{n}$
$N(\mu, \sigma^2)$	$(\sum X_i, \sum X_i^2)$		$\bar{X}$ and S^2	—	t / χ^2 tests
$U[0, \theta]$	$X_{(n)}$		$\frac{n+1}{n} X_{(n)}$	Pareto	reject $X_{(n)} > \theta_0 \alpha^{1/n}$
Cauchy(θ)	order stat. (not complete)		none in closed form	—	no MLR, no UMP

Useful facts. For $\text{Exp}(\theta)$, $\sum_i X_i \sim \text{Gamma}(n, \theta)$ and $\frac{2}{\theta} \sum_i X_i \sim \chi_{2n}^2$ — the standard route to an exact cutoff. For $U[0, \theta]$, $X_{(n)}$ has density nx^{n-1}/θ^n on $[0, \theta]$, so $\mathbb{E}X_{(n)} = \frac{n\theta}{n+1}$ and $\text{Var} X_{(n)} = \frac{n\theta^2}{(n+1)^2(n+2)}$. For $\text{Laplace}(0, 1)$ with density $\frac{1}{2}e^{-|x|}$: $\mathbb{P}(|X| > t) = e^{-t}$ for $t \geq 0$. $\mathbb{E}[X_1 | \bar{X} = t] = t$ and $\text{Var}(X_1 | \bar{X} = t) = \sigma^2(n-1)/n$ in the normal model.

14 One-page decision tree

Read the question, then follow the arrow

“Find the UMVUE of $\psi(\theta)$ ”

→ Is the family exponential? Identify T , $c(\theta)$, and check the natural parameter space contains an open interval $\Rightarrow S = \sum T(X_i)$ is complete sufficient. Then either match a series (Route A) or Rao–Blackwellise an indicator (Route B). If the support depends on θ , use the extreme order statistic and verify completeness directly.

“Does an unbiased estimator exist?”

→ Set $\mathbb{E}_\theta T = \psi(\theta)$ and expand. Binomial $\Rightarrow$ polynomial of degree $\leq n$. Poisson $\Rightarrow \psi(\theta)e^\theta$ analytic. Exponential $\Rightarrow$ Laplace transform uniqueness.

“Does it attain the CRLB?”

→ Compute $\partial_\theta \log L$; if it equals $k(\theta)[T - \psi(\theta)]$ then yes, and $\text{Var}(T) = \psi'(\theta)/k(\theta)$. Otherwise no. Check regularity first — moving support means the CRLB does not apply at all.

“Show A and B are independent”

→ Basu. Show A is complete sufficient and B is location- or scale-invariant, hence ancillary. (Or, in the normal case, use joint normality and zero covariance.)

“Find the Bayes estimate”

→ Posterior $\propto$ likelihood $\times$ prior; recognise the kernel; report the posterior mean. If the prior is improper, check the posterior is proper and report the generalized Bayes estimate.

“Show T is minimax”

→ Constant risk? Find a prior making T Bayes and quote the constant-risk theorem. Otherwise build π_k with $r(\pi_k) \rightarrow \sup_\theta R(\theta, T)$ and quote the limit theorem. If Θ is unbounded and Bayes risks blow up, conclude every estimator has infinite maximum risk.

“Find the MP test” (both hypotheses simple)

→ Neyman–Pearson. Form f_1/f_0 , find the shape of $\{\Lambda > k\}$, set the size, randomise if the distribution is discrete or if Λ has ties.

“Find the UMP test” (one-sided)

→ Establish MLR (exponential family with monotone $c(\theta)$, or a direct computation), then quote Karlin–Rubin and set the cutoff from the null distribution at the boundary θ_0 .

“Find the UMP test” (two-sided)

→ Normally none exists; say why (the one-sided best tests point in opposite directions). *Unless* the support moves with θ , in which case rejecting on the impossible-under- H_0 region is free and a UMP test can exist.

Sources: lecture notes 1–10 (29 Jul — 28 Aug 2026) and Homeworks 1–3.

Companion document: `PI_Assignment_Solutions.pdf` — worked solutions to all 15 homework problems.